



Philosophical and Linguistic Intuitions and the Core-Periphery Distinction



Friederike Moltmann

Introduction

Intuitions play an important role both in philosophy, or at least in metaphilosophical discussions about philosophical theorizing, ~~as well as~~ in linguistics. The aim of this ~~paper~~ is to illuminate the connection between philosophical and linguistic intuitions. Philosophical intuitions are a subject of controversy in current philosophical debate; yet many philosophers take intuitions to be essential for philosophical research: intuitions serve as philosophical evidence and as premises in philosophical arguments, and in cases of apparent conflicting intuitions, the basis of philosophical puzzles and paradoxes. Linguistic intuitions, which concern the acceptability of sentences or linguistic constructions, are considered the basis (if perhaps not the only basis) for linguistic theory, at least within a generative linguistic background.

There is an obvious connection between certain philosophical intuitions and linguistic intuitions, philosophers use ~~of~~ linguistic data in order to bring forth or clarify philosophical intuitions. Appeal to natural language in that way can be found in various philosophical domains, epistemology, philosophy of mind, philosophy of language, and especially metaphysics.

The ~~paper~~ will argue for the following two points. First, only certain parts of language or language uses can serve for uncovering or sharpening philosophical intuitions, namely what I call the core of language (or language use) as opposed to its periphery (or its peripheral use). Throughout history, philosophers when appealing to natural language have been careful to respect such a core-periphery distinction. The core-periphery distinction imposes a limit on the extent to which philosophical intuitions are reflected in natural language. Though universally adopted implicitly, the distinction between core and periphery has never been made explicit. However, it roughly matches the distinction between ordinary and non-ordinary use in ordinary language philosophy.

Second, both philosophical and linguistic intuitions are to be distinguished from beliefs as well as reflective or unreflective judgment or acceptance. For those cases in which philosophical intuitions correlate with linguistic intuitions, there are no reasons to distinguish the two: the philosophical ~~notion~~ that are reflected in language are just part of the knowledge of language. There are (partial) philosophies that are implicit in natural language that are thus part

of the knowledge of language. In particular, the ontology reflected in natural language, with the correspondence of grammatical should be viewed as part of grammar itself.

There are some subsidiary points this paper will make. One of them is that intuitions ontologically are best conceived as kinds of attitudinal objects in the sense of Moltmann (2003, 2021, 2024), that is, as *sui generis* entities ontologically on a par with claims, beliefs, and decisions that are neither propositional attitudes (which are relations between agents to contents) nor propositions (which are mind-dependent content bearers that come with satisfaction conditions).

Another subsidiary point will be that natural language displays a richer range of what should be recognized as philosophically relevant linguistic intuitions, in tune with the development of theoretical linguistics, a fact that will bear on the notion of an intuition itself.

~~1~~ ~~Intuitions in philosophy~~

Many philosophers in the past and present have endorsed the value of intuitions for philosophical theorizing. Intuitions play a role in various philosophical domains: philosophy of language, epistemology, metaphysics, and ethics. Intuitions about counterfactual situations help distinguish directly referential terms from definite descriptions, Gettier examples distinguish knowledge from justified true belief, intuitions about Socrates and the singleton of Socrates bear on the notions of ontological dependence and essential truth, intuitions about what actions are more wrong than others play a role in ethics.

Intuitions have also faced significant philosophical controversy, raising, among other things, the question whether intuitions really act as evidence in philosophy. The status of intuitions has been challenged in particular in experimental philosophy, which considers intuitions the subject of empirical study, with their potential interpersonal and cross-cultural inconsistencies. I share the view of the importance of intuitions both for philosophy and linguistics while recognizing the second challenge for intuitions being, in principle, ~~subject~~ of empirical study.

There are a number of general features of intuitions. First of all, intuitions need not be factive: there is an intuition that there is a set for every property, but that intuition is not correct. Second, intuitions are not easy and immediate, despite a common assumption. Intuitions may have to be brought forward through well-constructed thought experiments (Bengson 2020). Another way, less discussed in the literature, is through appeal to natural language data, that is, through linguistic intuitions. The philosophical literature usually makes use of only fairly simple kinds of linguistic data, involving subject-predicate predicates. That, of course, needs to be revised in face of theoretical development of linguistics with its wealth of syntactic and semantic generalizations, typological research, and theoretical analyses.

There are various views about what philosophical intuitions are. One view is that they are beliefs of some sort. But it has been pointed out, having an intuition

that P it is neither necessary or sufficient to believe that p or even be disposed to believe that P, for example in cases of paradoxes (all propositions in a paradox are intuitions but cannot all be believed) and perceptual illusions (the illusion is the content of an intuition, but not necessarily a belief) (Pust 2019). Still some philosophers prefer to consider intuitions to just be judgments of a sort (Williamson 2007). But the notion of a judgment does not really capture what is characteristic of intuitions. If a judgment is arrived at through reflection or reasoning, it is clearly not an intuition, which acts as a premise, not a conclusion, for philosophical reasoning. If a judgment is an immediate judgment, it may just fail to be the kind of intuition that is not immediate. There are also proposals to consider intuitions (certain kinds of) dispositions to believe (Sosa 1998). But intuitions appear to be episodic and thus unlike dispositions (Pust 2019). An alternative view assimilates intuitions to perception, as attitudes of ‘seeming’. Perception contrasts with reasoning, yet still may require some effort for its identification. On yet another view of intuitions ~~are~~ considered *sui generis* attitudes of ‘intuiting’. (As we will see shortly, no verb, actually is needed for capturing the specific attitude related to intuitions: having an intuition arguably displays the attitude just as transparently as ‘having a belief’ or ‘making a judgment’ displays the attitudes of belief and of judgment, a point I will return to shortly.)

Philosophical intuitions concern philosophically relevant propositions. Like propositions, intuitions themselves need to be content bearers: they are taken to have the status of evidence in philosophy and to play the role of premises in philosophical arguments. Yet intuitions are not just propositions: as intuitions they contrast with belief, knowledge, and visual perception. As content bearers, intuitions cannot be propositional attitudes, which are relations of an agent to a content, and they cannot be pure contents either. Intuitions furthermore are not mental states. Intuitions are shareable, which mental states are not. Moreover, intuitions come with correctness and strength conditions, which mental states don’t, at least not as the sorts of things we refer to as mental state (??? *John’s mental state is true / correct / implies that it will rain*). Intuitions have correctness conditions that are tantamount to conditions of truth (‘The intuition that a set corresponds to every property is incorrect’), intuitions enter inferential relations (‘the intuition that p implies that q’), intuitions enter relations of aboutness to a subject matter (‘intuitions about knowledge’), and intuitions also come with a part structure based on partial content (an intuition may be only partly correct). In addition to correctness conditions amounting to truth conditions, intuitions come with conditions of being right, real, or genuine, properties which are gradable (an intuition can be ‘weak’ or ‘strong’).¹ An intuition being ‘right’ or ‘correct’ does not or does not just amount to its representing what is the case; rather it means that it is a ‘real’ intuition, an intuition that, as an intuition, can be corroborated by thought experiment or language (see below).

1. In my ‘intuition’, even *correct* when applied to intuitions can have the reading of genuineness, rather than truth. Some attitudinal objects display a norm of correctness pertaining to grounds for maintaining a propositional content rather than the truth of the propositional content, for example proofs, perceptions, understandings, and conclusions. Note that correctness conditions for beliefs

What sorts of objects could intuitions be if they are neither propositional attitudes (being content bearers), nor mental states (having correctness conditions), nor propositions (having conditions of genuineness)? There is an ontologically category I have recently argued for and elaborated that appears particularly well-suited for intuitions, and that is that of an attitudinal object (Moltmann 2003, 2014, 2024). Attitudinal objects are entities such as claims, judgments, beliefs, decisions, intentions, and requests. Attitudinal objects are mind-dependent bearers of satisfaction conditions whose part structure is ordered by partial content. They typically go with satisfaction predicates that can obtain neither of events nor of propositions, such as *be correct*, *be fulfilled*, *be complied with*, *be carried out*. Attitudinal objects may incorporate various features beyond their propositional content, such as direction of fit, the strength of belief, degree of justification, and factivity (knowledge). If intuitions come with *prima facie* justification, then this is precisely what is constitutive of intuitions as objects beside their propositional content. Intuitions need not be taken to depend on the psychological state of a particular agent; rather they may be viewed as kinds of attitudinal object, which are not dependent on a particular agent. Attitudinal objects generally come in kinds. John's belief that *p* is particular to John, but 'the belief that *p*' can be shared among John and Joe. The same holds for intuitions: 'John's intuition that *p*' is John's, but 'the intuition that *p*' can be shared among John and Joe. Thus, intuitions are best viewed as kinds of attitudinal objects

Attitude verbs often correlate with complex attitude predicates consisting of a light verb like *have* or *make* and a noun for an attitudinal object (*believe* – *have a belief*, *judge* – *make a judgment*). Sometimes the complex predicate is the only option, as in the case of *have an impression/impression that p* or, I fact, *have an intuition / the intuition that p*). If attitudinal objects are considered objects *sui generis*, not reducible to propositions or mental states, then propositional attitudes will consist in agents standing in relation (of 'having' or 'making', for example) to attitudinal objects, with *that*-clause complements giving the content of an attitudinal object.

2 ~~Intuitions in linguistics~~

Linguistic intuitions are much less controversial than philosophical intuitions, at least in the tradition of generative linguistics.² Linguistic intuitions (largely) make up the very basis of linguistic research, at least on a generative linguistic view. These are the intuitions of native speakers of the language in question (including of linguists who speak the same language). Linguistic intuitions are about the acceptability of sentences and constructions, about the identity and incompatibility meanings, and under what conditions sentences may be true or

are not of this sort, given what is reflected in the use of *correct* or *right* in application to belief. Beliefs are correct just in case they are true.

2. Linguists also make use of linguistic corpora, in addition to native speakers' judgments.

false and entail each other.³

Linguistic intuitions share crucial features with philosophical intuitions. One of them is that linguistic intuitions come in degrees. In linguistics, the sensitivity to intuitions and their varying degrees of strength has developed over time. Whereas at an early stage no distinctions were made between ungrammatical or semantically unacceptable sentences, at later stages linguistics became aware that grammatically and acceptability judgments are graded. Such a fine-tuning regarding the strength of linguistic intuitions has developed not only during the history of linguistics, but generally also takes place within the development of individual researchers in linguistics.

Like philosophical intuitions, linguistic intuitions are not easy and immediate: they require developing a sense for data with their varying degrees of acceptability, and sometimes recourse to comparison with other native speakers' intuitions and subsequent re-examination and possibly sharpening of one's own intuitions.

In the case of linguistic intuitions, the distinction between intuition and belief is entirely apparent: linguistic intuitions are not beliefs about the acceptability of sentences and their truth conditions. Linguistic intuitions are simply manifestations of the knowledge or possession of relevant parts of language. Sharpening of one's own linguistic intuitions means recognizing a part of one's native language. Unlike beliefs, linguistic intuitions cannot be withdrawn on the basis of reasoning.

Linguistic intuitions also obviously fail to have the status of *a priori* beliefs. Even if, as generative linguists assume, natural language involves a universal innate grammar, linguistic intuitions do not only pertain to the universal aspects, but also language-specific ones.

~~3 Philosophical intuitions reflected in natural lan-~~ ~~guage~~

Throughout history, at some times more than others, philosophers have appealed to natural language in support of a philosophical view or notion. Natural language has in particular been used as a way of uncovering or clarifying metaphysical intuitions, in the context of descriptive metaphysics⁴. A specific version of descriptive metaphysics is natural language ontology, the branch of metaphysics whose subject matter is the ontology reflected in natural language (Moltmann 2022). Natural language ontology has a narrower subject matter in that it gives priority to those intuitions that have a manifestation in natural language.

3. As Paul Postal has pointed out to me, linguistic intuitions may also be about rhyme, registers, and relative strength (e.g., *astounded* vs. *surprised*).

4. Descriptive metaphysics is the branch whose subject matter has been characterized as our shared conceptual scheme (Strawson 1959) or, less Kantian, as the ontology reflected in ~~our 'ordinary judgments'~~, as Fine (2006) calls them, that is in our metaphysically relevant intuitions.

There is a reason why the connection between linguistic and philosophical intuitions is particularly important in the area of metaphysics, and that is because it bears on the notion of an object itself. In natural language semantics, referential NPs are generally taken to stand for entities and quantificational NPs are generally taken to range over entities; verbs moreover are taken to **predicated** of entities. In addition to the notion of an object itself, numerous metaphysical notions are reflected in the grammatical structure and the lexicon of natural languages, including time, space, causation, as well as ontological categories of various sorts. Natural language in a way involves ontology throughout.

Philosophers have drawn on natural language also for other philosophical domains, such as epistemology, philosophy of mind, and philosophy of language. Using natural language in support of a philosophical notion or view does not mean pursuing language-dependent philosophical view or notion. Making use of linguistic intuitions simply serves to uncover and clarify philosophical intuitions, when those intuitions are not as such obvious, the main purpose of philosophers' use of linguistic data in support of a philosophical notion or view. Philosophical intuitions themselves are not about linguistic meaning or structure, but about (apparent) reality or concepts. Natural language simply serves to make those intuitions apparent.

Not all areas or issues in philosophy have a linguistic reflection. Philosophy of physics, **biology** and largely moral philosophy have little connection to language. There is of course folk biology and folk physics, which psychologists have been interested in. However, evidence for those domains does not involve language in the way of uncovering intuitions. The data for folk biology and folk physics are rather different than those for philosophical domains implicit in language, by including assertions, for example (see Section 6).

~~4 Philosophical domains implicit in natural language~~

In the following, we will see some examples of how linguistic intuitions bear on issues in different philosophical domains other than metaphysics.

In epistemology, philosophers have drawn on the semantic behavior of the verb *know*, which is of course relevant for the notion of knowledge as are verbs of perception such as *see*, *seem*, *appear*, *look*, *taste*, and *sound*. It is not just the lexical content of particular verbs that is relevant for issues in epistemology, but also syntactic constructions or choices of functional words, such as the difference between *know that* and *know how*, which reflects the difference between theoretical and practical knowledge.

Likewise philosophers of mind often draw on the semantic behavior of attitude verbs like *believe*, *intend*, *think*, **hope** and *decide* when discussing the nature of mental attitudes. With attitude reports, what matters is not just the lexical content of attitude verbs, but also the construction of clausal complementation that attitude verbs generally involve. In fact, particular linguistic views about the construction of complementation with *that*-clauses has greatly influenced views in both philosophy of mind and language. A common assumption among

philosophers has been that *that*-clauses have the status of referential terms and stand for propositions as abstract bearers of truth; moreover, attitude verbs express two-place relations between agents and propositions. Given the apparent logical form, attitudes thus are relations to propositions, as denoted by *that*-clauses. The nature of propositions is reflected in the apparent behavior of *that*-clauses. Philosophers thus held the view that attitude reports show that attitude are relations to positions and *that*-clauses and terms standing for propositions. Not all clauses are always considered the same in their semantic role for mental attitudes. Thus, different types of complement clauses (*that*-clauses, *wh*-clauses, *how*-clauses), mood, and verb form (infinitival and finite clauses) appear to reflect different mental attitudes. Mental states and events are also linguistically reflected in the semantics of verbs and nominalizations.

Notions of the philosophy of mind may be reflected in other ways in the semantics of natural language. Thus, simulation, an important notion in cognitive science, is arguably involved in the semantics of generic *one*, in the form of generic simulation (Moltmann 2010). Given that analysis, natural language provides specific support for the notion of simulation.

Natural language is helpful also when clarifying the concept of an intuition itself, as we will see.

Drawing on natural language is less common for philosophy of language when it comes to linguistic data in which the relevant notions appear themselves. Notions of philosophy of language are generally arrived at through reflection on language. They are not objects of philosophical intuitions, unlike metaphysical judgments and judgments about epistemology and philosophy of mind.

There are some exceptions, though. One of them is the notion of truth, though philosophical theories of truth do not necessarily aim to provide a notion as the basis for a full semantic theory of natural language. Thus, debate about correspondence, coherence, idealist and pragmatic theories of truth hardly bear on semantic theory. ~~Ye~~, natural language can provide significant clarifications about issues such as what the bearers of truth are. Philosophers generally focus on truth predicates applying to *that*-clauses, taking it for granted that *that*-clauses act as terms referring to propositions. But natural language treats beliefs, reports, ~~claim etc.~~ likewise as truth bearers (Section 3.2.).

There are cases where philosophers have drawn on natural language to clarify a theoretical notion of semantic theory. A recent case in point is D'Ambrosio's (2019) analysis of the verb *refer* as an intentional transitive verb, which he takes to provide support for internalist semantics.

One might also argue in favor of a particular theory of meaning on the basis of linguistic facts about the verb *mean*. What is striking is that the verb *mean* is not an extensional transitive verb relating an expression to an entity that is a meaning, but rather it applies to two pure quotes:



(1)(a) 'Rouge' means 'red'.

(b) 'Rouge' means the ~~concept / meaning / denotation~~ of 'red'.

This seemingly gives support for the view that meaning is not a relation

between expressions and objects of some sort that are meanings. But rather perhaps a view on which meaning relates to uses of an expression of concept. The object argument position is in fact not referential, taking expressions or concepts as arguments (the denotation of pure quotes in referential position), but rather non-referential not permitting substitution by an explicit expression-referring term:



(2) Rouge means the concept / the word ‘red’.

Such data potentially bear on the ontology of linguistics. There is a debate about the ontological status of words, whether ~~are words~~ primarily abstract objects (types) or tokens (or classes / instantiated kinds of tokens) (Wetzel 2009; Miller 2020). If quotations are primarily ~~non~~referential as in (2), they may plausibly be viewed as predicates of utterances (phatic objects), which supports the view a token-first view of words.

The way natural language reflects situations or possible worlds overtly might also be used in support for a semantic theory based on either situation or worlds. Thus, there are various expressions in English that serve to refer to situations (*in that ~~case / situation~~, under those circumstances, then*), but there do not seem ~~to~~ terms that refer to entire possible worlds (‘possible world’ is of course a technical term, which does not count).

Appeal to such facts would not serve to uncover philosophical intuitions as subject matter, but rather to show that natural language directly reflects a notion of the metalanguage. But the insight can be put differently, namely that a notion of the metalanguage matches a linguistically reflected philosophical intuition; that is, that there is a coincidence between a reflective notion and a linguistically manifest intuitive notion.

To summarize, linguistic intuitions have been used in various philosophical domains in order to uncover or sharpen philosophical intuitions about philosophical concepts. The aim has been either a purely descriptive one, to uncover a notion that we implicitly adopt, or to defend a reflective philosophical notion, showing that it is an intuitive notion.

~~5 Metaphysics and linguistic intuitions~~

~~5.1 The special status of metaphysics~~

A special case of a philosophical domain implicit in natural language is metaphysics. The connection of natural language to metaphysics is particularly important, more so than its connection to epistemology, philosophy of mind, and philosophy of language and natural language. The grammar of natural language itself appears tied to metaphysics, or at least a descriptive understanding of metaphysics. Thus, a very influential view is that the notion of a referential NP reflects the notion of an object (Frege’s dictum ~~‘An object is what a referential NP (‘name’) stands for’~~). The category of properties is generally taken to be

reflected, at least to an extent, in predicates of natural language. Certain ontological categories appear to correspond to syntactic categories: verbs generally describe events; adjectives generally describe qualities or tropes (modes). In addition, syntactic features such as tense, aspect, and mood obviously relate to metaphysics as do functional elements, such as copula verbs (and other light verbs) and modal auxiliaries.

The connection between natural language and metaphysics also presents novel challenges for the notion of an intuition. In the domain of metaphysics, there are particularly striking discrepancies between ontological beliefs with their possibility of revision and the sort of implicit acceptance reflected in language, to the extent that it is in fact much less clear how to apply the notion of a metaphysical intuition. Moreover, even within natural language, different ontologies may be reflected at different levels of ontology (the grammatical-conceptual divide, cf. Section 3.5.).

5.2 ~~Appeal to natural language in metaphysics~~

In the following, I will list some examples of a traditional sort of appeal to natural language in metaphysics as well as some examples where syntactic analysis plays a particular role.

Appealing to natural language in support of metaphysical intuitions, as was mentioned, has been a practice throughout the history of philosophy, as part of the pursuit of descriptive metaphysics.

There is a longstanding debate about the identity of the statue and the clay constituting it. On a parsimonious view on which entities occupying the same spatial region at the same time are identical, the state and the clay should be the same entity; on a more permissive ontological view, the state and the clay are distinct entities. Fine (2003) argued against the identity of the statue and the clay drawing on the understanding and applicability of predicates in natural language. Thus, (3a, 4a) and (3b, 4b) have different truth conditions and (5a) is acceptable, but not so (5b):

- (3)(a) The statue is nice.
- (b) The clay is nice.
- (4)(a) The statue is new.
- (b) The clay is new.
- (5)(a) The statue is tall.
- (b) ??? The clay is tall.



Here the understanding of predicates or their applicability to particular types of entities is used to clarify metaphysical intuitions about the distinction between material and objects constituted from it, and in fact the distinctive natures of the two types of entities.

Another example ~~again~~ is partly again from Fine's work, and that is that natural language supports the view of events as distinct in type from material objects. The choice of existence and spatial modifiers of existence predicates

display that distinction. Material objects go with the existence predicate *exist*; events are incompatible with *exist* and take as existence predicates *happen*, *occur*, *take place*, and *go on* instead:



- (6)(a) The protest is still going on.
- (b) The protest still exists.
- (7)(a) The destruction of the temple really occurred / ??? really existed.
- (b) The temple really existed / ??? occurred / ??? took place / ??? happened.

Moreover, spatial modifiers are possible with existence predicates for events, but not existence predicates for material objects:



- (8)(a) The vase still exists.
- (b) ??? The vase exists in that room.
- (c) The attack took place / happened / occurred in Paris.

Again, these differences point not ~~only~~ to an ontological distinction between events and material objects, but rather to a particular distinction among them regarding their relation to space and time (see Fine 2003, Moltmann 2020b). They also manifest a non-univocal notion of existence, distinguishing modes of being for events and for material objects.

Philosophers have also drawn on the applicability or reading of different predicates to clarify the ontological events and facts (Vendler 1967). Thus, temporal and causal predicates are applicable to events, but not facts:



- (9)(a) The speech lasted two hours / triggered an applause.
- (b) ?? The fact that John spoke lasted two hours / triggered an applause.

Linguistic data have also served the purpose of uncovering an as yet unrecognized ontological category. An example is Twardowski's (1999) category of products as part of his distinction between actions and products. Twardowski argued for a distinction between mental acts or speech acts (acts of thinking, acts of deciding, acts of claiming, acts ~~or~~ requesting) and their enduring 'products', judgments, decisions, claims, requests. The linguistic data he took to display the distinction were two sorts of nominalizations in Polish, German, and French. In English, these correspond to gerunds (or explicit act-referring terms) and simple nominalizations. Twardowski showed that different sorts of predicates apply to the two kinds of nouns; in particular only nouns for products allow for predicates of truth or satisfaction, not nouns for actions:



- (10)(a) John's claim was true.
- (b) ?? The act of claiming was true.
- (11)(a) John's judgment was correct.
- (b) ?? The act of judging was correct.
- (12)(a) The request was fulfilled.



- (b) ?? The act of requesting was fulfilled.
 (13)(a) The decision was carried out.
 (b) ??? The act of deciding was carried out.

The action-product distinction, while not entirely new at the time, is certainly not a distinction that belongs to the standard ontological repertoire. Natural language here served to put forward an ontological category that is not generally recognized as such. For Twardowski, the linguistic data show that it is products that are truth bearers, not abstract propositions, as on standard views. Drawing on a distinction in natural language allowed him to uncover a philosophical intuition that most philosophers are not likely to even be able to articulate.

Not only can natural language serve the purpose of distinguishing between philosophical preconceptions and philosophically relevant intuitions, it can also serve the purpose of uncovering philosophically relevant intuitions in the first place when those could hardly have been articulated otherwise.

5.3 ~~Philosophically relevant intuitions and syntactic construction~~

The way philosophers have traditionally drawn on natural language generally involves simple subject-predicate sentences (and perhaps sometimes the use of quantifiers and pronouns). Given the development of linguistics as a contemporary discipline, it is important to recognize that linguistic data are now the subject matter of highly developed empirical and theoretical fields of study. This means that for the purpose of clarifying philosophically relevant intuitions, the use of linguistic data should in principle be systematic, taking into account the full range of linguistic facts, including syntactic constructions as analysed within a particular syntactic theory and crosslinguistic generalizations.

Here are just two examples where syntactic research has become particularly relevant in the use of natural language sentences in support of philosophical intuitions. The first example concerns the ontology of numbers. Frege (1884) argued for numbers being objects on the basis of apparent referential occurrences of number words and functional number terms in apparent identity statements such as:

(14) The number of planets is eight.

Frege took referential NPs themselves to be indicative of objecthood, with his slogan ~~'an object is what a referential NP (or 'name') may stand for'~~. ~~'Being able to flank the identity symbol'~~ for him was one of the criteria for referential NPs – and he took *be* to express identity in sentences like (14). Given these assumptions, *the number of planets* and *eight* in (14) come out as terms referring to numbers as objects. The apparent intuition that numbers are objects was thus supported by the occurrence of apparent number-referring terms in sentences like (14).

Frege's argument, however, is problematic in view of more recent syntactic

research. Ever since Higgins (1973) there has been a different take on the syntactic structure of sentences like (14), namely on which they are not identity statements involving object-referring terms, but specificational sentences, which plausibly convey a relation between a question and an answer, in the case of (14) roughly ‘how many planets are there?’ – ‘there are eight planets’. Alternatively, sentences like (14) have been taken to convey an identity among higher-order semantic values, including that of quantifiers like *eight*, again providing no evidence for number as objects.⁵

Another important example of the involvement of syntactic analysis to support philosophical intuitions is *that*-clauses. Most philosophers took it for granted that *that*-clause complements of attitude verbs like *believe* act as referential terms denoting propositions. The linguistic form of attitude reports appears to display propositions as entities that are shareable truth bearers and objects of attitudes. However, there is a different emerging view, namely that *that*-clauses are not referential terms referring to propositions, but act as predicates of content-bearers *such* attitudinal objects (Section 4) a view that has both syntactic and semantic motivations (Moulton 2015; Moltmann 2014, 2022, 2024).

If metaphysical intuitions are uncovered or clarified through linguistic facts (in the broad sense), then such metaphysical intuitions are in fact on a par with linguistic intuitions. In particular, metaphysically relevant intuitions reflected in linguistic data are about the ontology that is part of the semantics of natural language, at least on the standard view of semantics. In that sense metaphysical intuitions reflected in natural language go together with semantic intuitions, and *as intuitions* can hardly have a different status than the linguistic intuitions themselves. The knowledge and use of a natural language goes along both with syntactic and semantic intuitions about that language as well as the philosophically relevant intuitions reflected in the core of that language.

These examples should suffice to indicate that the realm of linguistic data to consider in the interest of clarifying or uncovering philosophical intuitions has significantly expanded, involving not only semantics, but also syntactic data of various sorts, against the background of syntactic analyses within particular syntactic theories.

~~6 The linguistic data relevant for philosophical domains reflected in natural language and the core-periphery distinction~~

There is a limit as to what sorts of linguistic data can *be* and have been used to uncover or clarify philosophical intuitions. This limit relates to the close connection between linguistic intuitions and philosophical intuitions that have a linguistic reflection. Natural language can be drawn on only to uncover philo-

5. See Moltmann (2013) for discussion and further references.

sophical notions that are implicit in natural language, that is, that automatically go along with the use of language. It cannot serve to support philosophical notions that can be described, upon reflection, by using natural language. That is, not all of language (or language use) is suited to uncover or clarify philosophical intuitions. Natural language can be used to convey philosophical reflection, but that is an entirely different way in which natural language reflects philosophical views of notions.

How should the two uses or parts of language be distinguished, use of language to describe philosophical judgments or notions arrived at through reflection and use of language that by itself comes with particular philosophical views or notions and thus reflects philosophical intuitions? Here are some criteria that distinguish them, which hold not only for metaphysics, but any philosophical domain

First of all, philosophical assertions cannot be considered indicative of philosophical intuitions. Philosophical assertions may be part of folk metaphysics, the ‘naïve’ metaphysics adopted by non-philosophers (or folk philosophy of mind, folk epistemology, etc.); but this is not the sort of linguistic data philosophers draw on in support of metaphysical intuitions. Of course, philosophical assertions may have as their content the content of philosophical intuitions, but philosophical assertions are not considered linguistic support for philosophical intuitions. In fact, they are never considered evidence. Thus, no philosopher could appeal to (15) in support of an ontology of events or of abstract objects:



- (15)(a) There are events.
- (b) ??? Abstract objects exist.

Nor could a philosopher appeal to denials of (15a, b) to argue for the opposite.

There are instead other data that are indicative of an ontology of events implicit in natural language. These are not so much referential NPs describing events such as *John's walk* or *the second world war*. While it is uncontested that referential NPs can refer to events, a more compelling linguistic support for an ontology of events implicit in natural language has been the semantic behavior of adverbials. This is what led Davidson to posit events as implicit arguments of verbs, as in the logical form in (16b) for (16a), where adverbials are regarded predicates of events:



- (16)(a) John buttered the toast with a knife at midnight.
- (b) $\exists e(\text{butter}(e, \text{John}, \text{he toast}) \ \& \ \text{with a knife}(e) \ \& \ \text{at midnight}(e))$

What is also generally excluded when drawing on language to support metaphysical intuitions is technical expressions and philosophical terms such as *essence*, *ontological dependence*, *the fundamental*, *the part-of-relation*, *possible world*, *sum*, *proposition*, *existence*, *property*, *object*, *group*. Likewise, ‘non-ordinary’, philosophical uses of ~~uses of~~ expressions need to be set apart (the kinds of uses of expressions ordinary language philosophers objected to, cf. Section 8). The parts of language or its use that can legitimately be drawn on when clarifying philosophical intuitions can be called the ‘core of language’ and the parts or use

of language that are avoided the ‘periphery of language’.⁶

The exclusion of certain parts of language or uses of languages is ~~the~~ matter of metaphilosophical observation about the way philosophers draw on linguistic expressions or uses of expressions in support of philosophical intuitions. The core-periphery distinction will ultimately have to be explained, but this is a task separate from the metaphilosophical observation itself.

A few clarifications are needed regarding the core-periphery distinction. What is peripheral is not meant to be statistically rare or less important. Rather the notions of core and periphery are simply meant to capture the distinction between implicit acceptance (in first approximation) and reflective acceptance. Roughly, uses of expressions in the periphery require some degree of philosophical reflection, but not so uses of expressions in the core, which instead may display philosophical intuitions.

How exactly is the core-periphery distinction to be drawn? As was already said, technical expressions belong to the periphery, as do sortals, whose content does constitute a degree of philosophical reflection. Furthermore, non-ordinary, philosophical uses of language belong to the periphery (though not all of language ~~permits~~ non-ordinary, an issue I will turn to shortly). In addition, certain constructions by themselves belong to the periphery. In particular these are what I call reifying terms, certain sorts of close appositions, as below:

- 
- (17)(a) the truth value true
(b) the number eight
(c) the proposition that it might rain
(d) the event of raining
(e) the sum of John and Mary

The exclusion of reifying terms from the data that can be used in support ~~of~~ philosophical is widely attested in the relevant literature. For example, the existence of the term *the truth value* true was not used by Frege (1918/9) to motivate truth values as objects. Rather Frege’s motivation was distinguishing sense and reference for clauses in a way that parallels the distinction for referential NPs. Frege (1884), moreover, did not motivate numbers as objects on the basis of terms like *the number eight*, but on the basis of *the number of planets* and *eight*. His argument for numbers as objects could in principle have been much easier: *the number eight* is a referential term and it can hardly refer to anything but a number. But Frege knew that a term like *the number eight* could not make the argument. Instead he used sentences like *the number of planets is eight*, which involves only terms from the core of language, *eight* and *the number of planets*, terms that won’t involve reflection.

Another example concerns the linguistic motivations for propositions. Many philosophers have appealed to natural language to motivate propositions, which have become an important notion for philosophy of language, philosophy of mind, and ontology. However, philosophers, hardly ever motivate propositions

6. See Moltmann 2020aa, 2022 as well as already (Moltmann (2013)).

on the basis of the existence of NPs like *the proposition that it is raining* in natural language.⁷ Rather propositions are motivated by the apparent function of *that*-clauses as referential terms and the apparent role of *that*-clauses of providing arguments for attitude verbs like *believe*, generally considered two-place predicates expressing relations between agents and propositions.

Davidson's (1967) linguistic motivations for an ontology of events did not come from terms like *the event of raining*, but on the basis of inference patterns with adverbials. In fact, it is the semantics of verbs and adverbial modifications that is generally considered grounds for considering events an important part of the ontology of natural language, not event sortals.

Link (1983) and many other semanticists following him have argued that (unrestricted) sums of individuals form part of the ontology of natural language. But they did not motivate sums of individuals on the basis of terms like *the sum of John and Mary*, but rather on the basis of the semantic behavior of plurals and conjunctions (*the students, John and Mary*), expressions clearly from the core of language.

What these observations indicate is that philosophers and semanticists have always made tacit use of a distinction between core and periphery of language (or language use): only the core, not the periphery, reflects the ontology of natural language. Reifying terms, like technical terms and technical uses of terms, belong to the periphery, not the core of language.

~~7 The 'content' of the core-periphery distinction~~

What characterizes expressions or uses of expressions that make up the periphery of language? In a first characterization, expressions or uses of expressions in the periphery of language involve some degree of philosophical *reflection*, whereas the core of language is tied to implicit acceptance. That is, the kind of contrast involved is that between implicit 'acceptance' as opposed to inferential acceptance, reasoning, or reflection. Reflection appears to go along with the structure of reifying terms themselves, that is, with their compositional semantics. Let us assume that the general structure of reifying terms is as below (Moltmann 2013, chap. 7):

 (18) ~~the-~~ sortal noun – non-referential material

~~Non~~referential material will include *that*-clauses, number words, plural terms, and quotations.

Reifying terms then involve an operation of abstraction based on the content of non-referential material. When viewed as representing a mental act, such an operation of introduction ~~tion~~ an object by abstraction should count as a form of reflection.

7. There are some exceptions, for example Schiffer (2003).

The core-periphery distinction does not correspond to a distinction in the nature of the ontology or philosophical concepts themselves. A philosophical notion that serves as the intended meaning of a peripheral (use of an) expression (based on reasoning) may turn out to be part of the core as well. Rather what matters is a distinction between a form of implicit acceptance (in first approximation) and reflective acceptance. For a philosophical domain (philosophical notions or an ontology) that is reflected in the core language the following condition, preliminarily, holds:

-  (19) A philosophical domain implicit in natural language is a philosophical domain a speaker *implicitly accepts* by way of using the *core* of the language.

Nouns by themselves, it seems, always permit a peripheral use. The noun *event* is used by philosophers and semanticists as conveying a concept that is considerably broader than that conveyed by the noun *event* on an ordinary use (where it is restricted to telic events and distinguished from actions and states). Likewise a noun like *property* on its own can be used by different philosophers to convey different conceptions of a property. Non-ordinary uses have been subject to attack by ordinary language philosophers (Ayer, Wittgenstein, Ryle). Ordinary language philosophers argued that philosophical problems arise precisely from non-ordinary, philosophical uses of language, thus rejecting philosophically reflective uses of language. However, non-ordinary uses are not all illegitimate linguistically: language is designed so as to permit non-ordinary, philosophical uses, through extension of meaning, introduction of technical terms, and ‘conceptual engineering’.

That said, as a matter of fact, there are limits of non-ordinary uses of language. Certainly, functional expressions and syntactic features do not permit a non-ordinary use. There is no such thing as a philosophical use of a plural, say on which the formation of pluralities is restricted by a particular philosopher’s condition. Plural formation as well as NP-conjunction *applies* in an entirely unrestricted fashion (*the things that were not mentioned, Quine and the Eiffel Tower*). In fact, capturing the ordinary (and only) meaning of functional expressions such as the conditional, the definite determiner, quantifiers, and negation is the subject of philosophical research itself. The same does not hold for the semantics of nouns like *property*, *individual*, and *predicate*, terms from the periphery of language.

The core-periphery distinction does not entirely match the functional-lexical divide, though. Not all lexical words permit a non-ordinary use, or so it seems. In contrast to nouns, verbs, it seems, generally do not permit a non-ordinary use. This contrast between nouns and verbs is apparent from the way the verb *exist* and the noun *existence* can be used. Many philosophers take existence to be a univocal notion applying to anything there is, and they can use the noun *existence* to convey the univocal notion. Other philosophers may restrict existence to the mode of being of material and abstract objects and can use *existence* accordingly. The verb *exist*, however, cannot be used to convey the univocal notion of *exist*: it strictly resists an application to events,



- (20)(a) ??? The rain still exists now.
(b) ??? The protest existed yesterday.

When using the verb *exist*, rejection of the non-univocal notion of existence (as endurance) conveyed by *exists* in favor of a univocal notion of existence is impossible. Non-ordinary use seems to be nouns by themselves: while *existence* can be used to convey a univocal notion, *the existence of the rain*, *the existence of the protest*, and *the existence of the accident* are rather unacceptable.

Thus, while the lexical-functional distinction plays a role, there are other parameters at stake as well. Obviously further work needs to be done on how the core-periphery distinction is ~~anchored~~ grammar, and an explanation would need to be developed why the distinction obtains in the way it does.

There is also an interesting issue of how the core-periphery distinction relates to Chomsky's (1981; 1986) distinction in syntax with the same name. Given Chomsky's distinction in syntax, very roughly, the core of the syntactic system of a language represents universal grammar and the periphery exceptions and outside influences. Chomsky's distinction obviously has completely different motivations than the present core-periphery distinction. Let us just note that on that distinction the syntactic structure of reifying terms should belong to the core and thus a coincidence of the two distinctions is at least not obvious. Yet, for both distinctions the lexical-functional divide plays a role (Yang 2016).

8 ~~The status of the core of language~~

Only the core, not the periphery of language (or language use), reflects philosophically relevant intuitions, in the sense that the core is associated with a form of implicit acceptance of notions in a particular philosophical domain, rather than acceptance arrived at through reasoning or reflection. How should this notion of implicit acceptance be understood?

One important observation is that the kind of implicit acceptance associated with the core of language does not permit rejection. Even when a speaker rejects, upon reflection, a philosophically relevant notion or category conveyed by part of the core of language, she will automatically be committed to it again when she uses that part of the language. The core of natural language does not permit non-ordinary, philosophical uses.

The impossibility of rejection is not tied to the *implicitness* of the relevant form of acceptance. Implicit acceptance when understood as a weak form of belief (or disposition to believe) does allow for rejection. For example, implicit acceptance in the form of bias in ethics allows for rejection and subsequent revision of belief. The kind of implicit acceptance of philosophical notion or views implicit in natural language just is not belief, not even a weak or dispositional form of belief. The fact that philosophically relevant intuitions reflected in natural language resist rejection adds to the other arguments put forward in the literature that intuitions are distinct from beliefs.

With its impossibility of rejection, the implicit acceptance of philosophical notion or views implicit in the core of language is on a par with linguistic intuitions. Linguistic intuitions, provided they are correct, obviously do not allow for rejection. Linguistic intuitions rather come with the knowledge of the language in question (at least given a generative linguistic background). This means that linguistically reflected philosophically relevant intuitions are on a par with knowledge of language. Knowledge of the language goes along with the possession of philosophically relevant intuitions that are reflected in the core of language. One may even go further and consider such intuitions part of the knowledge of grammar itself, if that notion is suitably extended.

As with philosophically relevant intuitions in general, this raises the question whether intuitions reflected in natural language could be mistaken or could be compatible with alternative notions or views. The answer will of course depend on the extent to which philosophy should be guided by intuition. It may also depend on whether one would adopt a permissive conception of the relevant philosophical domain, on which, for example, the ontology that is reflected in natural language could be just one ontology beside other ontologies, which may have been accepted through reasoning.

9 Conclusions

Throughout history philosophers have drawn on language to clarify or uncover philosophically relevant intuitions. They did so while making use of an implicit distinction between the core and the periphery of natural language: only the core, not the periphery, can be used to clarify or uncover such intuitions. Philosophically relevant intuitions that are reflected in the core of a natural language do not have the status of beliefs or even weaker forms of acceptance. Rather they have the very same status as linguistic intuitions. The possession of both sorts of intuitions comes with the knowledge and use of the core of natural language. By contrast, the periphery of language (or language use) is suited to convey philosophical notions based on reflective acceptance. The contrast intuition—reflection thus has a correlate in the architecture of language itself.

Acknowledgements

I would like to thank Gabriel Dupre, Paul Boghossian, and Paul Postal for comments on an earlier version of this paper. The paper has also benefitted from the audiences at Philosophy of /and Linguistics Conference at Keele University in June 2023, the symposium Philosophy of Linguistics during SOPHiA, Salzburg, September 2023, and the conference Philosophy and Generative Grammar 2 at the Universidad del Valle, Columbia, May 2024.

References

- Bengson, J. 2020. "The Myth of Quick and Easy Intuitions." In *Routledge Handbook of Linguistic Reference*. New York: Routledge.
- Chomsky, N. 1981. *Lectures on Government and Binding: The Pisa Lectures*. Dordrecht: Foris.
- . 1986. *Knowledge of Language: Its Nature, Origin, and Use*. Westport, CT and London: Praeger.
- D'Ambrosio, J. 2019. "Semantic Verbs are Intensional Transitives." *Mind* 128 (509): 213–248.
- Davidson, D. 1967. "The Logical Form of Action Sentences." In *The Logic of Decision and Action*, edited by N. Rescher, 81–95. Pittsburgh: Pittsburgh University Press.
- Fine, K. 2003. "The Non-Identity of a Material Thing and Its Matter." *Mind* 112:195–234.
- . 2006. "In Defense of Three-Dimensionalism." *Journal of Philosophy* 103 (12): 699–714.
- . 2017. "Naïve Metaphysics." *Philosophical Issues* 27.
- Frege, G. 1918/9. "Thoughts." In *Collected Papers on Mathematics, Logic, and Philosophy*, edited by B. McGuinness, 351–372. Oxford: Blackwell.
- . 1884. *Die Grundlagen der Arithmetik*. N/A: N/A.
- Higgins, F. R. 1973. "The pseudo-cleft construction in English." Published in 1979 by Indiana University Linguistics Club. PhD diss., MIT.
- Link, G. 1983. "The Logical Analysis of Plurals and Mass Terms: A Lattice-theoretical Approach." In *Meaning, Use and Interpretation of Language*, edited by R. Bäuerle et al., 303–323. Berlin: De Gruyter.
- Miller, J. 2020. "The Ontology of Words: Realism, Nominalism, and Eliminativism." *Philosophy Compass* 15 (7): 1–13.
- Moltmann, F. 2013. *Abstract Objects and the Semantics of Natural Language*. Oxford: Oxford University Press.
- . 2014. "Propositions, Attitudinal Objects, and the Distinction between Actions and Products." *Canadian Journal of Philosophy* 43 (5–6): 679–701.
- . 2020a. "Abstract Objects and the Core-Periphery Distinction in the Ontological and Conceptual Domain of Natural Language." In *Abstract Objects: For and Against*, edited by J. L. Falguera and C. Martínez, 255–276. Dordrecht: Springer.
- . 2020b. "Existence Predicates." *Synthese* 197 (1): 311–335.

- Moltmann, F. 2021. "Truth Predicates, Truth Bearers, and their Variants." *Synthese* 198:689–716.
- . 2022. "Natural Language Ontology." <https://plato.stanford.edu/archives/fall2022/entries/natural-language-ontology/>.
- Moulton, K. 2015. "CPs: Copies and Compositionality." *Linguistic Inquiry* 46:305–342.
- Pust, J. 2019. "Intuition." <https://plato.stanford.edu/archives/sum2019/entries/intuition/>.
- Schiffer, S. 2003. *The Things We Mean*. New York: Oxford University Press.
- Sosa, E. 1998. "Minimal Intuition." In *Rethinking Intuitions: The Psychology of Intuitions and Its Role in Philosophical Inquiry*, edited by M. DePaul and W. A. Ramsey. Lanham, MD: Rowman / Littlefield.
- Strawson, P. 1959. *Individuals: An Essay in Descriptive Metaphysics*. London: Methuen.
- Twardowski, K. 1999. "Actions and Products: Some Remarks on the Borderline of Psychology, Grammar, and Logic." In *Kazimierz Twardowski: On Actions, Products, and Other Topics in the Philosophy*, edited by J. Brandl and J. Wolenski, 103–132. Amsterdam and Atlanta: Rodopi.
- Vendler, Z. 1967. *Linguistics in Philosophy*. Ithaca, NY: Cornell University Press.
- Wetzel, L. 2009. *Types and Tokens: On Abstract Objects*. Cambridge, Mass.: MIT Press.
- Williamson, T. 2007. *The Philosophy of Philosophy*. New York: Routledge.
- Yang, C. 2016. *The Price of Linguistic Productivity: How Children Learn to Break the Rules of Language*. Cambridge, MA: MIT Press.